Subject: Fundamentals of Engineering Mathematics -II

Module 1 : Differential Equations of First Order and First Degree

Linear Differential Equations

First Order first Degree Linear Differential Equations:
Consider

$$\frac{dy}{dx} + P(x)y = Q(x) \qquad \frac{dx}{dy} + P'(y)x = Q'(y)$$

where P and Q are functions of x alone or constants.
and P' and Q' are functions of y alone or constants.
Such differential equations are called **Linear** Differential Equations.
Examples: Linear
1. $\frac{dy}{dx} + 3xy = x^2$
2. $\frac{dy}{dx} + y = e^x$

Examples: non-Linear
1. $\frac{dy}{dx} + xy = x^2 + y^2$
2. $\frac{dy}{dx} + xy^2 = 4$

Method for solving Linear Differential Equation (LDE)(Proof for reference only - will not be asked in the exam)

LDE: $\frac{dy}{dx} + P(x)y = Q(x)$
$\therefore (Py - Q)dx + dy = 0$———(1)
On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M = (Py - Q) \qquad & and \qquad N = 1 \\ \therefore \frac{\partial M}{\partial y} = P \qquad & \qquad \frac{\partial N}{\partial x} = 0 \\ \therefore \frac{\partial M}{\partial y} & \neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

$$\begin{aligned} \therefore \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N} &= \frac{P - 0}{1} \\ &= P(x) \\ &= f(x)(\text{function of } x \text{ only}) \end{aligned}$$

$\therefore I.F. = e^{\int f(x)dx} = e^{\int P(x)}dx$

Multiplying the given DE by this I.F. we have,

$$(Py - Q)e^{\int P(x)dx} + e^{\int P(x)}dx = c$$

Hence the general solution is given by

$$\begin{aligned}\int Mdx + \int \text{(terms in N free from } x) \ dy =& c\\ \Rightarrow \int (Py - Q)e^{\int P(x)dx}dx + \int 0 \ dy =& c\\ \Rightarrow \int Pye^{\int P(x)dx}dx =& \int Qe^{\int P(x)dx}dx + c\\ \therefore y \int Pe^{\int P(x)dx} =& \int Qe^{\int P(x)dx}dx + c\end{aligned}$$

Consider $y \int Pe^{\int P(x)dx}$.

$$\begin{aligned}Put\ w &= e^{\int P(x)dx}\\ \Rightarrow \frac{dw}{dx} &= e^{\int P(x)dx}\ \frac{d}{dx}\int P(x)dx\\ \Rightarrow \frac{dw}{dx} &= e^{\int P(x)dx}\ P(x)\\ dw &= e^{\int P(x)dx}\ P(x)\\ \Rightarrow y\int Pe^{\int P(x)dx} &= y\int\ dw\\ \Rightarrow y\int Pe^{\int P(x)dx} &= yw =\end{aligned}$$

Therefore the general solution is given by

$$\begin{aligned}ye^{\int P(x)dx} &= \int Qe^{\int P(x)dx}dx + c\\ i.e\ y\ \textbf{(I.F.)} &= \int Q\ \textbf{(I.F.)}dx + c\end{aligned}$$

where **I.F.**=$e^{\int P(x)dx}$

Note:-The general solution of the linear DE of form

$$\frac{dx}{dy} + P'(y)x = Q'(y) \qquad \text{is}$$

$$x(\mathbf{I.F.}) = \int Q'(y)(\mathbf{I.F.})dy + c - - - where\ \mathbf{I.F.} = e^{\int P'(y)dy}$$

Examples:

1. Solve $x \ \log x \ \dfrac{dy}{dx} + y = 2\log x$

   Solution: The given DE can be written as

$$\frac{dy}{dx} + \frac{y}{x\log x} = \frac{2}{x}$$

The given DE is in the form $\dfrac{dy}{dx} + P(x)y = Q(x)$
where $P(x) = \dfrac{1}{x\log x} \quad \& \quad Q(x) = \dfrac{2}{x}$
Thus the given DE is linear.

Hence it's general solution is:

$$y(I.F.) = \int Q \cdot (I.F.)dx + c$$

where $I.F. = e^{\int P(x)dx} = e^{\int \frac{1}{x\log x} \ dx} = e^{\int \frac{1/x}{\log x} \ dx} = e^{\log(\log x)} = \log x$
Hence the general solution is:

$$\begin{aligned}
y(I.F.) =& \int Q \cdot (I.F.)dx + c \\
y(\log x) =& \int \frac{2}{x} \log x \ dx + c \\
y(\log x) =& 2\int \frac{1}{x} \log x \ dx + c \\
\text{putting} \quad \log x =& t \Rightarrow \frac{1}{x}dx = dt \\
\therefore y(\log x) =& 2\int t \ dt + c \\
\therefore y(\log x) =& 2\frac{t^2}{2} + c \\
\therefore y(\log x) =& (\log x)^2 + c
\end{aligned}$$

2. Solve $\left[\dfrac{e^{-2\sqrt{x}}}{\sqrt{x}} - \dfrac{y}{\sqrt{x}}\right]\dfrac{dx}{dy} = 1$

   Solution: The given DE can be written as

$$\frac{dy}{dx} = \left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right]$$
$$\frac{dy}{dx} + \frac{y}{\sqrt{x}} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

The given DE is in the form $\dfrac{dy}{dx} + P(x)y = Q(x)$
where $P(x) = \dfrac{1}{\sqrt{x}}$ & $Q(x) = \dfrac{e^{-2\sqrt{x}}}{\sqrt{x}}$
Thus the given DE is linear.

Hence it's general solution is:

$$y(I.F.) = \int Q \cdot (I.F.)dx + c$$

where $I.F. = e^{\int P(x)dx} = e^{\int \frac{1}{\sqrt{x}}\ dx} = e^{2\sqrt{x}}$
Hence the general solution is:

$$\begin{aligned}
y(I.F.) &= \int Q \cdot (I.F.)dx + c \\
ye^{2\sqrt{x}} &= \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}} e^{2\sqrt{x}}\ dx + c \\
ye^{2\sqrt{x}} &= \int \frac{1}{\sqrt{x}}\ dx + c \\
\therefore ye^{2\sqrt{x}} &= 2\sqrt{x} + c
\end{aligned}$$

3. Solve $(1 + x + xy^2)dy + (y + y^3)dx = 0$

Solution: The given DE can be written as $(1 + x + xy^2)dy = -(y + y^3)dx$

$\therefore \dfrac{(1 + x + xy^2)}{-(y + y^3)} = \dfrac{dx}{dy}$

$\therefore \dfrac{1}{-(y + y^3)} + \dfrac{(x + xy^2)}{-(y + y^3)} = \dfrac{dx}{dy}$

$\therefore \dfrac{dx}{dy} = \dfrac{(x(1 + y^2))}{-y(1 + y^2)} + \dfrac{1}{-(y + y^3)}$

$\therefore \dfrac{dx}{dy} = \dfrac{x}{-y} + \dfrac{1}{-(y + y^3)}$

$\therefore \dfrac{dx}{dy} + \dfrac{x}{y} = -\dfrac{1}{y + y^3}$

The given DE is in the form $\dfrac{dx}{dy} + P'(y)x = Q'(y)$
where $P'(y) = \dfrac{1}{y}$ & $Q'(y) = -\dfrac{1}{y + y^3}$
Thus the given DE is linear.

Hence it's general solution is:

$$x(I.F.) = \int Q' \cdot (I.F.)dy + c$$

where $I.F. = e^{\int P'dy} = e^{\int \frac{1}{y}\, dy} = e^{\log y} = y$
Hence the general solution is:

$$\begin{aligned} x(I.F.) &= \int Q' \cdot (I.F.)dy + c \\ x \cdot y &= \int -\frac{1}{y+y^3} \cdot ydy + c \\ x \cdot y &= \int -\frac{1}{1+y^2} \cdot dy + c \\ x \cdot y &= -tan^{-1}y + C \end{aligned}$$

Practice Problems:

1. $\sin 2x \dfrac{dy}{dx} = y + \tan x$
2. $(y+1)dx + (x - (y+2)e^y)dy = 0$

Equations reducible to linear equation form

Form1: $f'(y)\frac{dy}{dx} + P(x)f(y) = Q(x)$ Form2: $f'(x)\frac{dx}{dy} + P'(y)f(x) = Q'(y)$
put $f(y) = v$ in form 1 & $f(x) = v$ in form 2

These substitutions will transform Differential equations of above type to linear form. This can be illustrated with the help of following examples.

1. Solve $\dfrac{dy}{dx} + \left(2x\tan^{-1}y - x^3\right)(1+y^2) = 0$
Solution: The given DE can be written as

$$\frac{1}{(1+y^2)}\frac{dy}{dx} + 2x\tan^{-1}y = x^3$$

put $\tan^{-1}y = v$
on differentiating this w.r.t. 'x' we get
$\dfrac{1}{(1+y^2)}\dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$\therefore \frac{dv}{dx} + 2x \cdot v = x^3$$

This is linear equation in 'v'.
Here $P(x) = 2x$ and $Q(x) = x^3$

Hence it's general solution is:

$$v(I.F.) = \int Q \cdot (I.F.)dx + c$$

where $I.F. = e^{\int Pdx} = e^{\int 2x} = e^{x^2}$
Hence the general solution is:

$$
\begin{aligned}
v(I.F.) &= \int Q \cdot (I.F.) dx + c \\
v \cdot e^{x^2} &= \int x^3 \cdot e^{x^2} dx + c \\
\text{put } x^2 &= t \Rightarrow 2x\ dx = dt \\
\therefore v \cdot e^{x^2} &= \int \frac{t}{2} e^t dt + c \\
\therefore v \cdot e^{x^2} &= \frac{1}{2} \int t e^t dt + c \\
&= \frac{1}{2} [t \int e^t dt - t' \int \int e^t dt] + c \\
&= \frac{1}{2} [t e^t - (1) e^t] + c \\
\therefore v \cdot e^{x^2} &= \frac{1}{2} [x^2 e^{x^2} - e^{x^2}] + c \\
\therefore \tan^{-1} y \cdot e^{x^2} &= \frac{1}{2} [x^2 e^{x^2} - e^{x^2}] + c
\end{aligned}
$$

2. $\dfrac{dy}{dx} + x \cdot \sin 2y = x^3 \cos^2 y$

Solution: The given DE can be written as

$$
\begin{aligned}
\frac{1}{\cos^2 y} \frac{dy}{dx} + \frac{x \cdot \sin 2y}{\cos^2 y} &= x^3 \\
\frac{1}{\cos^2 y} \frac{dy}{dx} + \frac{x \cdot 2 \sin y \cos y}{\cos^2 y} &= x^3 \\
sec^2 y \frac{dy}{dx} + 2x \tan y &= x^3
\end{aligned}
$$

put $\tan y = v$

on differentiating this w.r.t. 'x' we get

$sec^2 y \dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$
\therefore \frac{dv}{dx} + 2x \cdot v = x^3
$$

This is linear equation in 'v'.

Here $P(x) = 2x$ and $Q(x) = x^3$

Hence it's general solution is:

$$v(I.F.) = \int Q \cdot (I.F.)dx + c$$

where $I.F. = e^{\int Pdx} = e^{\int 2x} = e^{x^2}$
Hence the general solution is:

$$\begin{aligned}
v(I.F.) &= \int Q \cdot (I.F.)dx + c \\
v \cdot e^{x^2} &= \int x^3 \cdot e^{x^2} dx + c \\
\text{put } x^2 &= t \Rightarrow 2x\ dx = dt \\
\therefore v \cdot e^{x^2} &= \int \frac{t}{2} e^t dt + c \\
\therefore v \cdot e^{x^2} &= \frac{1}{2} \int te^t dt + c \\
&= \frac{1}{2}[te^t dt - (1)e^t dt] + c \\
&= \frac{1}{2}[te^t - e^t] + c \\
\therefore v \cdot e^{x^2} &= \frac{1}{2}[x^2 e^{x^2} - e^{x^2}] + c \\
\therefore \tan y \cdot e^{x^2} &= \frac{1}{2}[x^2 e^{x^2} - e^{x^2}] + c
\end{aligned}$$

Practice Problems:

1. $\sin 2x \dfrac{dy}{dx} = y + \tan x$
2. $(y+1)dx + (x - (y+2)e^y)dy = 0$
3. $dr + (2r\cos\theta + \sin 2\theta)d\theta = 0$
4. $\dfrac{dy}{dx} + \dfrac{4x}{x^2+1} y = \dfrac{1}{(x^2+1)^3}$
5. $y\dfrac{dx}{dy} = x - yx^2 \sin y$